



POLITECNICO  
MILANO 1863

DEPARTMENT  
OF AEROSPACE SCIENCE  
AND TECHNOLOGY

FLY MI

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# CONCEPT PAPER

13 - Fly-Mi - Politecnico di Milano



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## ACADEMIC LEAD

Lorenzo Trainelli

TEAM LEADER

ACADEMIC LEAD



# 1 Introduction

We are a team of full-time students from *Politecnico di Milano*, united by our passion for Unmanned Aerial Vehicles (UAVs). Each year, we design and fly a new UAV from scratch, striving to innovate and build upon the achievements of previous projects. This year, as a first step in the UAS Challenge, we present our Concept Paper, which aims to provide a comprehensive yet concise overview of the preliminary design we have developed for our aircraft, outlining the rationale behind our choices and presenting the overall project plan and key technical drawings. This paper’s abbreviations and nomenclature will align as closely as possible with those outlined in the UAS Challenge guidelines. The new terminology is explained in the following section.

## 2 Nomenclature

LG *Landing Gear*

MTOM *Maximum Take-Off Mass*

PL *Payload*

PLDM *Payload Drop Mechanism*

## 3 Project Plan

For project planning, efforts proceeded on multiple fronts simultaneously. A detailed analysis of expenses from the previous project was conducted, leading to a budget and business plan for the current project to guide sponsor outreach. In parallel, a comprehensive Gantt chart was created to account for each phase of the project, potential errors and delays, and key submission dates required by the competition (Figure 4). At this stage, a work breakdown structure and resource allocation matrices were also developed to optimize task organization.

## 4 Conceptual Design

### 4.1 Weight Estimation

The design process began with a detailed analysis of the competition requirements and scenarios. The first objective was to determine the UAV’s MTOM. With the size of the aircraft limited by the team’s experience and the level of design detail and manufacturing’s available time limited by the competition schedule, a 7 kg MTOM was selected as the optimal starting point to balance project risks and competition performance. This estimation is based on data from previous projects, statistical analysis, and data from OEM-published products. The following table summarizes a preliminary weight distribution:

| Element            | Mass [kg] |
|--------------------|-----------|
| Electronics + PLDM | 0.50      |
| Main battery       | 0.60      |
| Motor + Prop       | 0.34      |
| Structure          | 3.32      |
| PL                 | 2.24      |
| MTOM               | 7.00      |

Table 1: Mass Estimation

Regarding materials, composites and additive manufacturing have been identified as the best solution to achieve airworthiness while ensuring improved system integration and structural optimization. Moreover, these techniques enable us to research innovative, eco-friendly solutions with our partners.

### 4.2 Electronics

The selected firmware for autonomous flight is Ardupilot, an open-source platform known for its robust community support and frequent updates. The main sensors include LiDAR for precise takeoff and landing, a barometer, dual IMUs, an airspeed sensor for wind correction, and dual GPS units for added safety. Hardware selection prioritises range, reliability, and safety. For this reason, 2.4GHz communications are paired with ExpressLRS protocol and mLRS for ground control, using high-gain antennas and a diversity receiver. The ESC will support telemetry and reverse thrust for landing. Eventually, all components will

undergo rigorous testing to ensure optimal performance. The first flight testing campaign was conducted this October. The initial system showed promising results in maintaining the predetermined path and velocity, even in windy conditions. Future improvements will focus on increasing precision in velocity and altitude maintenance, as well as implementing the take-off and landing phases.

### 4.3 Propulsion

Research and testing were conducted to identify an optimal propulsion system solution. In particular, various types of batteries were tested, and those with the best energy density (J/kg), power density (W/kg), and discharge performance (nominal voltage) were selected. Additionally, linear regressions were developed to link battery weight and total internal energy. Further research was made to match different motors with OEM propeller performance data.

Last year's propellers wind tunnel testing campaign was also exploited for result validation.

### 4.4 Payload Drop Mechanism (PLDM)

The system designed for payload release includes two guides that restrict its movement to vertical sliding. Inside the aircraft, it is supported by a wire wrapped around both the payload and the locking mechanism, which consists of a servo-actuated pin and pin housing mounted in the fuselage. When ready to release, the pin retracts, freeing the wire and allowing the payload to drop from the payload bay. Once outside the aircraft, a fabric strip deploys the parachute, which then detaches. The payload bay is then closed by a spring-loaded door that opens under the weight of the payload and automatically closes with the spring's force.

### 4.5 Preliminary Sizing

The preliminary sizing of the aircraft was conducted using the *Sizing Matrix Plot (SMP)* technique. The tool provides a robust basis for trade-off analysis and matching several performance requirements. Using MTOM and landing weight as inputs, the SMP determines the desired wing loading and power loading (Figure 1).<sup>1</sup>

To ensure reliable estimations, the aircraft's polar curves (takeoff, cruise, landing) and the optimal wing aspect ratio (AR) for structural integrity and aerodynamic performance, were based on data from our two previous UAVs.

In conclusion, a performance forecast and sensitivity analysis were conducted to visualize and quantify the impact of some design choices on the competition score and requirements. Furthermore, the expected maximum range of the aircraft was evaluated by varying the PL mass (Figure 2).

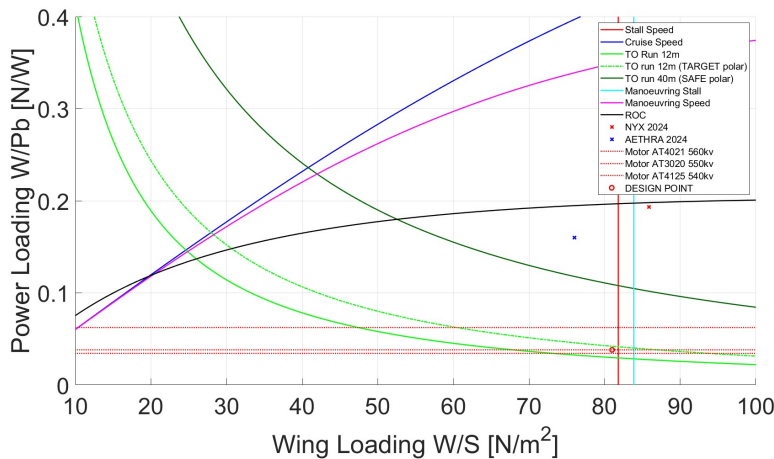


Figure 1: SMP

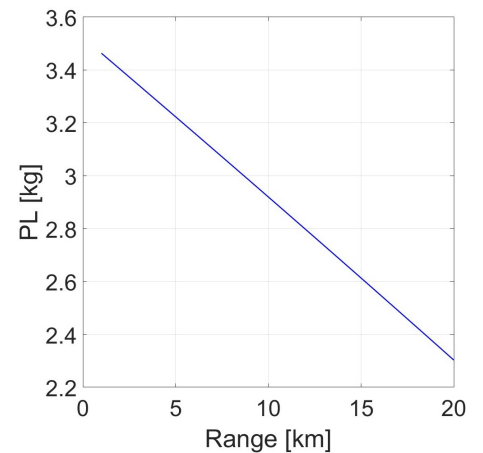


Figure 2: PL mass vs Range

<sup>1</sup>Figure 1 also shows possible choices of off-the-shelf motors as horizontal dotted lines

|                                  | Safe  | Optimal |
|----------------------------------|-------|---------|
| Wing Loading [N/m <sup>2</sup> ] | 81.0  | 81.8    |
| Power Loading [N/W]              | 0.038 | 0.042   |

Figure 3: Selected design point

## 4.6 Configuration

After extensive discussions and brainstorming on aerodynamics, stability, control, propeller clearance, and weight considerations, the team converged to a final aircraft configuration: **High-wing** design paired with a **V-Tail** and a **Tricycle LG**. The rationale behind this choice includes:

- **Wing and Fuselage Design:** The wing positioning selection involved several considerations. Both Mid-wing and parasol-wing solutions were excluded due to poor fuselage integration, the Mid-wing limits PL positioning and the parasol-wing is significantly less structurally efficient. High-wing and Low-wing configurations were therefore investigated. Aerodynamic and stability trade-off studies were carried out and two concepts were compared:

1. High-wing with PL inside the fuselage.
2. Low-wing with a minimal fuselage (housing only for electronic components) and externally placed payload.

The results indicated a significant increase in aerodynamic drag with the second option, with few differences in terms of control, ease of manufacturing, and airworthiness between the two. Consequently, since the drag is the main distinction factor, first option was chosen.

- **Tricycle LG:** The tricycle LG was preferred over a Taildragger LG to guarantee the best ground control and stability, reduced risk for the autonomous system, durability for harsh landings, good ground clearance during the take-off rotation, and the possibility to safely implement reverse thrust during landing. However, this choice introduces drawbacks such as increased weight, aerodynamic drag, and a more challenging structural integration.
- **V-Tail Configuration:** The V-tail design has been identified as the best compromise between weight savings and drag reduction, while still maintaining good stability and control characteristics. The wing's downwash on the V-tail was analyzed and deemed manageable. A 3D-printed V-tail will be produced during the second planned testing campaign (Figure 4) and mounted on Nyx (2024 project), enabling further acquisition of knowledge regarding both manufacturing and testing of the V-tail solution.
- **Payload Positioning:** The PL is placed coincident with the empty aircraft's centre of gravity. This ensures stability during all flight phases and makes the PL drop manoeuvre as safe as possible, even in the event of a system failure.

## 5 Safety and Sustainability

Sustainability will be addressed by adapting industry practices to our project, including the implementation of a life cycle assessment to understand and reduce the UAV's environmental impact. This will involve collaborating with companies that test advanced techniques (such as recyclable moulds), reusing waste materials, and taking measures to offset the drone's environmental impact. Regarding safety, we will identify risks, analyze their impact and probability, and use risk matrices for categorization. Human factors will be assessed, past safety issues reviewed, and targeted protocols and procedures developed to minimize risks and reduce individual responsibility in the event of an incident.

## 6 Project Planning

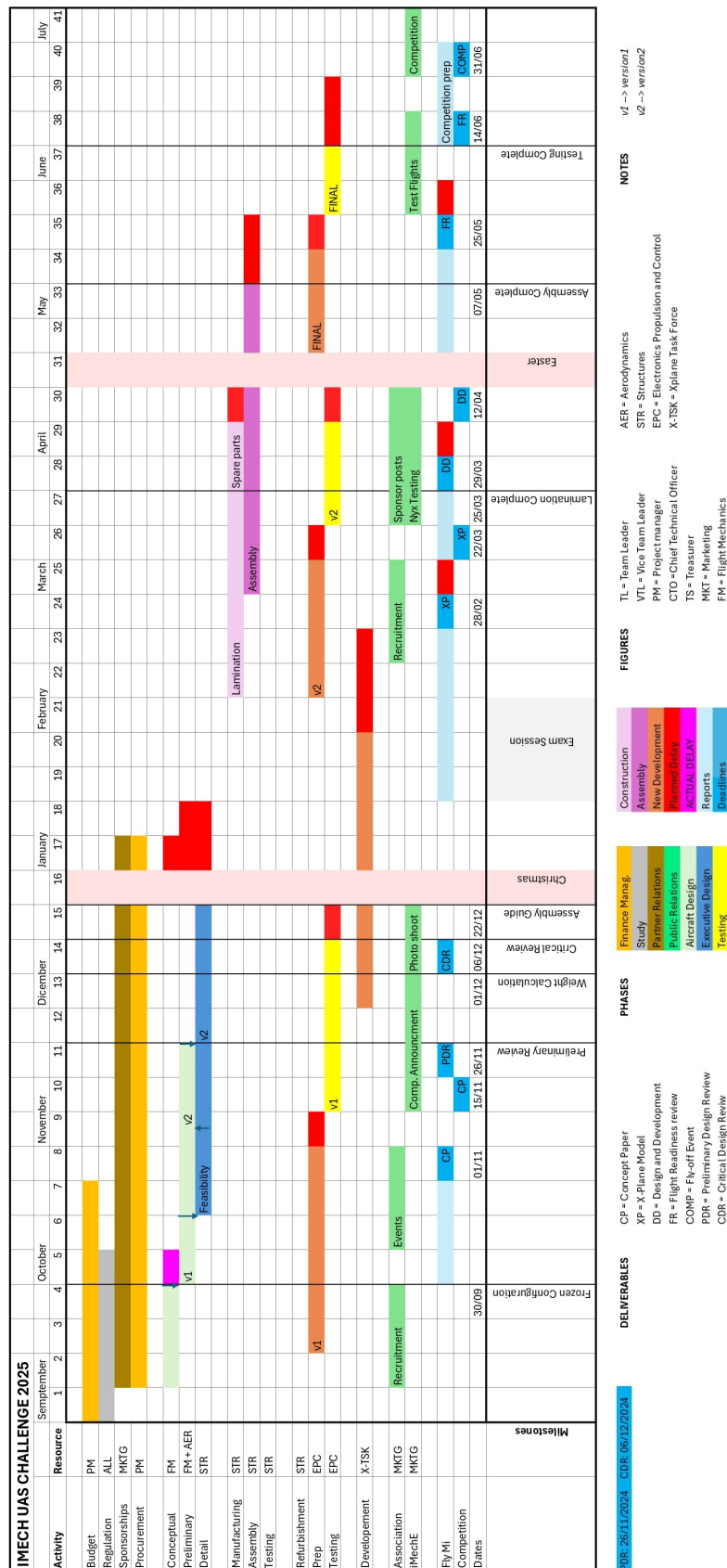
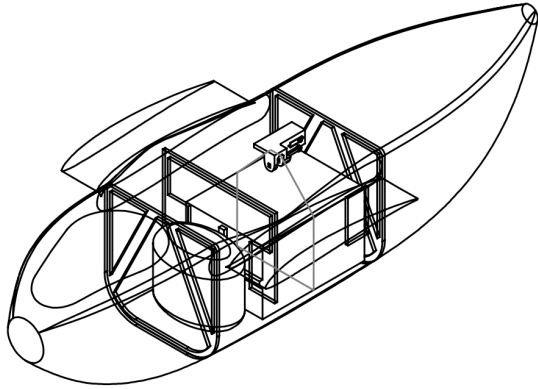
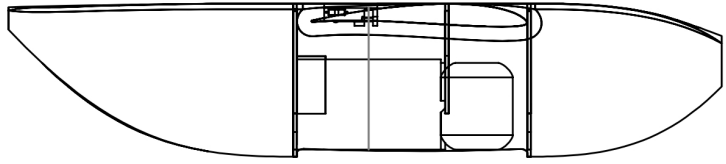
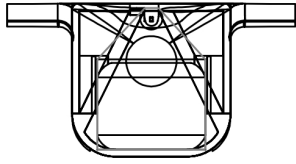
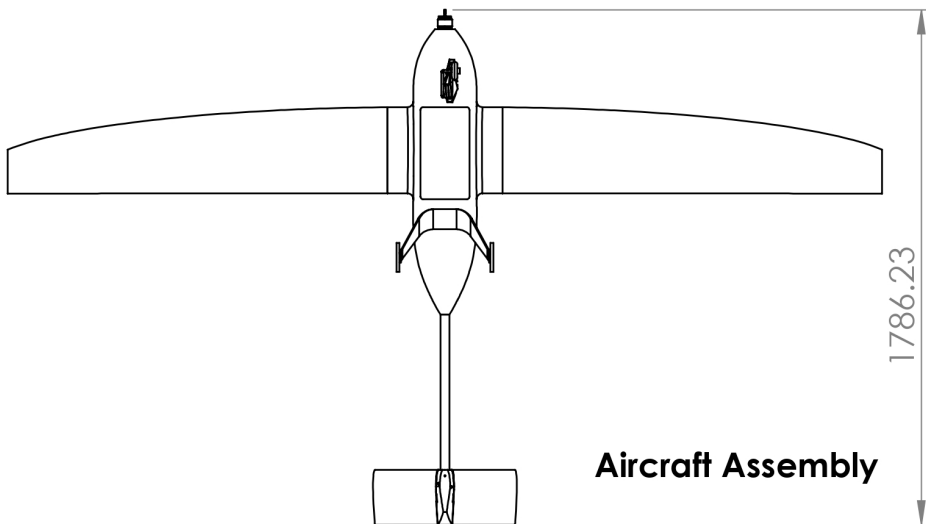
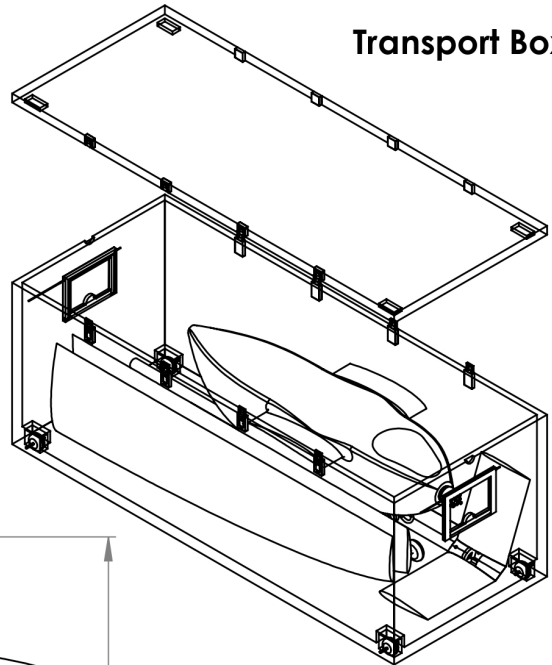


Figure 4: Project GANTT Chart

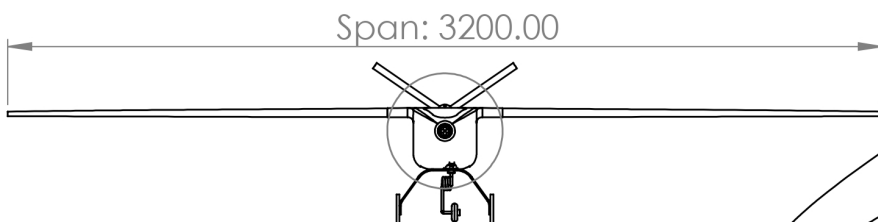
# Payload Release System



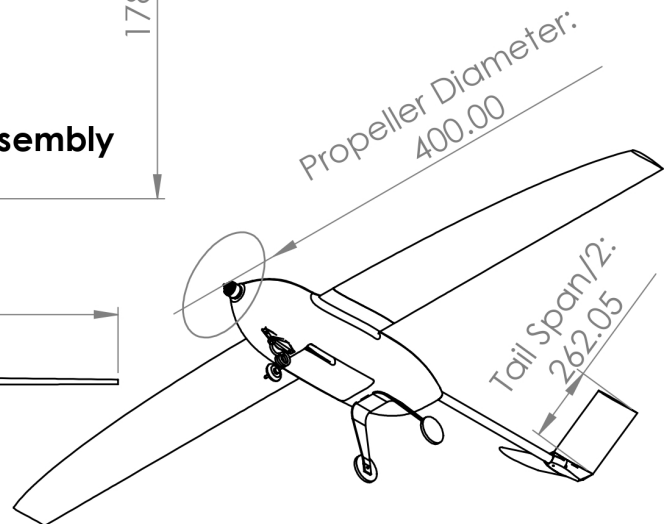
**Transport Box**



**Aircraft Assembly**



Span: 3200.00



Propeller Diameter:  
400.00

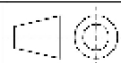
Tail Span/2:  
262.05

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A4

IMECHE **UAS Challenge**

**SCALE:** Aircraft: 1:25, Payload Release: 1:10, Box: 1:20

## Concept TECHNICAL DRAWINGS

all measurements are in mm